

Green - Controller
Pink - Class
Blue - Logic/Rules
Yellow - Enum
Red - UI

All UI talks
ONLY to the
controller

GameTableUI talks
PlayerHand,
OpponentHand, Deck,
DiscardPile, Buy,
MeldArea

Controller

GameController

- game : Game

- rules : RulesEngine

+ startGame()

+ startRound()

+ handleDraw(player, card)

+ handleDiscard(player, card)

+ handleLayDown(player, melds)

+ handleLayOff(player, card, meld)

+ handleBuy(player)

+ endTurn()

+ endRound()

UML class diagrams

Summarize

Customize

Comment

Record video

Game

- players : List<Player>

- deck : Deck

- discardPile : DiscardPile

- currentRound : Round

- currentPlayerIndex : int

+ nextPlayer()

+ getCurrentPlayer()

Round

- roundNumber : int

- requirement : RoundRequirement

- cardsPerPlayer : int

+ isComplete()

Player

- hand : Hand

- melds : List<Meld>

- score : int

- buysUsed : int

- hasLaidDown : boolean

+ drawCard(card)

+ discard(card)

+ layDown(melds)

+ layOff(card, meld)

Hand

- cards : List<Card>

+ addCard(card)

+ removeCard(card)

+ getScore()

Card

- suit : Suit

- rank : Rank

- isWild : boolean

+ getValue()

DiscardPile

- cards : Stack<Card>

+ top()

+ discard(card)

+ draw()

Meld (abstract class)

- cards : Stack<Card>

+ isValid()

Deck

- cards : Stack<Card>

+ shuffle()

+ draw()

RulesEngine

+ validateSet(cards)

+ validateRun(cards)

+ validateLayDown(player, melds, round)

+ canBuy(player)

+ isWild(card)

+ calculateScore(hand)

+ canLayOff(card, meld)

enums

SUIT

CLUBS
DIAMONDS
HEARTS
SPADES

enums

RANK

ACE
TWO
THREE
FOUR
FIVE
SIX
SEVEN
EIGHT
NINE
JACK
QUEEN
KING
JOKER

enums

ACTION TYPE

DRAW
DISCARD
BUY
LAY_DOWN
LAY_OFF

enums

ROUND REQUIREMENT

TWO_SETS
(sets=2, runs=0)
ONE_SET_ONE_RUN
(sets=1, runs=1)
TWO_RUNS
(sets=0, runs=2)
THREE_SETS
(sets=3, runs=0)
TWO_SETS_ONE_RUN
(sets=2, runs=1)
TWO_RUNS_ONE_SET
(sets=1, runs=2)
THREE_RUNS
(sets=0, runs=3)
THREE_SETS_ONE_RUN
(sets=3, runs=1)
THREE_RUNS_ONE_SET
(sets=1, runs=3)
FOUR_SETS
(sets=4, runs=0)
FOUR_RUNS
(sets=0, runs=4)

GameUI

- controller : GameController

+ showScreen(screenType)

+ showError(message)

+ refresh()

HomeScreenUI

+ showHallOfFame()

+ onLoginClicked()

LoginScreenUI

+ submitLogin(username, password)

+ onForgotPassword()

+ onGameCodeEntered(code)

GameTableUI

- playerViews : List<PlayerHandUI>

- discardView : DiscardPileUI

- deckView : DeckUI

+ renderTable()

+ updateTurn(player)

+ showBuyOption()

+ showRules()

PlayerHandUI

- cards : List<CardUI>

+ renderHand()

+ selectCard(card)

+ discardSelected()

+ attemptLayDown()

OpponentHandUI (face-done)

- cardCount : int

+ renderBacks()

Discard Pile UI (center)

- topCard : CardUI

+ renderTop()

+ onCardClicked()

Buy Button UI

+ requestBuy()

Meld Area UI (laid-down sets & runs)

- melds : List<MeldUI>

+ renderMelds()

+ highlightLayOffTargets()

Rules Popup UI

+ showRules()

+ close()

• ON PLAYER OWNS
EXACTLY one hand
• CREATES/OWNS melds once laid down